

Zixun Huang

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RESEARCH INTERESTS

Theoretical foundations of machine learning, with an emphasis on **Optimization Theory & Scaling Laws**, **LLM Reasoning & Post-Training & Agentic RL**, and **applications of AI**.

EDUCATION

University of Pennsylvania, Wharton School <i>Ph.D. in Statistics and Data Science</i>	Aug. 2026 – Present
Peking University, School of Mathematical Sciences <i>B.S. in Statistics</i> ; Elite Undergraduate Program for Applied Mathematics; GPA: 3.7/4.0	Sept. 2022 – Jun. 2026
University of California, Berkeley <i>Visiting Student</i> ; GPA: 4.0/4.0	Jan. 2025 – Aug. 2025

WORKING EXPERIENCE

Tencent HY – Qingyun Talent Program , <i>Research Intern</i>	Jun. 2026 – Aug. 2026
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PUBLICATIONS

- [1] Jia-Nan Wang*, **Zixun Huang***, Kairui Li*, Lei Wu[†]. *Momentum in Large-Batch Training: Polyak Enlarges the Critical Batch Size, Nesterov Improves Data Efficiency*. **Preprint**.
- [2] **Zixun Huang***, Kishan Panaganti*, Haitao Mi, Leowei Liang. *FlowSD: Trajectory-Balanced On-Policy Self-Distillation*. **Preprint**.
- [3] Zhongzhi Li*, Yucheng Shi*, Zongxia Li*, Ruhan Wang, Anhao Li, **Zixun Huang**, Junyao Yang, Lei Ke, Ninghao Liu, Haitao Mi, Leowei Liang. *Recursive Synthesis for Long-Horizon Terminal Tasks*. **Preprint**. [\[Paper\]](#) [\[Hugging Face\]](#) [\[Website\]](#)
- [4] **Zixun Huang***, Jiayi Sheng*, Zeyu Zheng. *Variance-Aware Baselines and Adaptive Learning Rates for Reinforcement Learning with Verifiable Rewards*. **Under review**. [\[Paper\]](#)
- [5] Yuhao Sun, Zekun Wu, **Zixun Huang**, Peijie Zhou. *TracingFlow: A Simulation-Free Trajectory Inference Framework Based on Second-Order Dynamics*. **Under review**.
- [6] Binghui Li*, Fengling Chen*, **Zixun Huang***, Lean Wang*, Lei Wu. *Functional Scaling Laws in Kernel Regression: Loss Dynamics and Learning Rate Schedules*. **NeurIPS 2025 Spotlight**. [\[Paper\]](#) * Equal contribution.

EXPERIENCE

Optimization Theory & Scaling Laws

- **Peking University**, *Research Intern*, Beijing May. 2023 – Aug. 2026
Supervisor: Assistant Professor Lei Wu
- **Large-batch momentum [1]**: Theoretically characterized how Polyak and Nesterov momentum reshape stochastic-training dynamics; established that Polyak enlarges the critical batch size, while Nesterov improves data efficiency.
 - **Functional scaling laws [6]**: Developed an intrinsic-time analysis of one-pass SGD that predicts full loss trajectories under arbitrary learning-rate schedules; validated the theory on power-law kernels and 0.1B–1B language-model pre-training.

LLM Reasoning & Post-Training & Agentic RL

- **Tencent Hy LLM – Qingyun Talent Program**, *Research Intern*, Beijing Apr. 2026 – Present
Supervisor: Senior Researcher Kishan Panaganti
- **On-policy self-distillation [2]**: Developed FlowSD, a trajectory-balance-based OPSD algorithm for learning distributions over complete responses. Across five reasoning benchmarks, improved over GRPO by **1.95** and **2.12 points** on Qwen3-4B and Qwen3-8B, while increasing semantic strategy diversity among successful solutions.
 - **Long-horizon agent training [3]**: Built a recursive synthesis pipeline that evolves verified terminal tasks into harder executable tasks, producing **37,484 tasks** across 15 rounds at approximately \$0.05 per task; the resulting trajectories improved Qwen3.5 agents through SFT and RL.
- **University of California, Berkeley**, *Research Intern*, Berkeley, CA Jan. 2025 – Aug. 2025
Supervisor: Associate Professor Zeyu Zheng
- **Variance-aware policy optimization [4]**: Derived a variance-optimal baseline and an SNR-adaptive learning-rate rule for KL-regularized RLVR with convergence guarantees. On Qwen3-4B-Base, OBLR-PO improved average Pass@1 over GRPO by **2.52 points** across four benchmarks.

SELECTED HONORS AND AWARDS

Grand Prize, Challenge Cup Competition	2026
Weiming Scholar, Peking University	2026
Hong Sheng Scholarship	2024
First Prize, Chinese Mathematics Contest	2024
Yau Contest Groups Prize	2024
Xiaomi Scholarship	2023
Gold Prize, Chinese Mathematical Olympiad	2021
Silver Prize, Chinese Mathematical Olympiad	2020

SKILLS

Programming: Python (NumPy, pandas, scikit-learn, PyTorch), C++, MATLAB, L^AT_EX

Language: GRE V160/Q170; GRE Math 970 (95th percentile); TOEFL 101